

A LIGHTWEIGHT AND MULTI-SCALE CNN MODEL FOR LAND-COVER CLASSIFICATION WITH HIGH-RESOLUTION REMOTE SENSING IMAGES

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ABSTRACT

Accurate and timely landcover information plays an important role in land resources management and urban planning. In this article, a lightweight and multi-scale convolutional neural network model is proposed for high-resolution remote sensing images classification. Its inputs are the multi-scale patches to capture the multi-scale variability of spatial features in high-resolution remote sensing images. In this model, some standard convolution is replaced by depth separable convolution, which has fewer parameters and requires less calculation cost. The experiments were conducted on high-resolution images of two different regions (i.e., Beijing and Qingdao, China). The promising performance verified the proposed method is very efficient for the classification of high-resolution remote sensing imagery.

Index Terms—landcover classification, HRS imagery, CNN, lightweight, multi-scale

1. INTRODUCTION

In recent years, more and more high-resolution remote sensing (HRS) images have become available. The HRS imagery provides abundant spatial information, and it is widely used in many fields, such as resources management [1], urban planning [2], and environment science [3]. Generally, land-cover mapping is the basis of these applications. However, the accurate and timely classification remains a challenging task, due to the rich details and extremely diverse spectral signals of objects emerging in HRS images [4]. To solve these problems, a wide range of traditional methods for land cover classification of HRS images have been proposed [5-7]. These methods categorize images based on different spatial units, including pixels, moving windows, objects, and scenes [8]. However, the traditional methods only consider the shallow manual features, such as spectral, texture features, it is difficult to achieve optimal classification accuracy.

Comparatively, deep learning can automatically exploit the hierarchical features from input data to describe the complex landcover structures or patterns [9]. As a typical

deep learning model, CNN is powerful to model spatial relationships between images, making them have a strong capability to depict high-level and semantic features of images. Besides, CNN has much fewer connections and parameters due to its weight sharing and local connection scheme, which makes it easier to train. Another advantage of CNN is the use of pooling operations, which results in the CNN learning slightly translational-and rotational-invariant features [10], which is a desirable property for complex land cover patterns.

Since these advantages, CNN has received extensive attention in HRS images classification [8, 11, 12]. Despite of the improved performance, there are two main problems in applying CNNs to the land-cover classification of HRS images. Firstly, mostly the CNNs (e.g. VGG, ResNet) still have many parameters due to their large number of convolution layers, which may hamper the accuracy of the classifier when the number of available labeled training samples is limited. Secondly, the land cover class distribution of HRS image is complicated, traditional CNN with fixed-size windows as input is not flexible enough. Convolution with different spatial context size may capture more discriminative context feature for HRS classification [13].

Inspired by the above analysis, we designed a lightweight CNN, using the deep separation convolution with fewer parameters and faster computation. Besides, three different scale patches are employed as inputs for the CNN to learn multi-scale features to improve the classification accuracy and robustness of the model. Therefore the experiments were conducted on two regions (i.e., Beijing and Qingdao) using GaoFen2 multispectral images with 4 m resolution. The results demonstrated the reliability of the proposed model.

2. MATERIALS AND METHODS

In this current research work, we developed a lightweight CNN by combining multi-scale inputs and deep separation convolution. The detailed structure of the CNN model is shown in Fig.1.

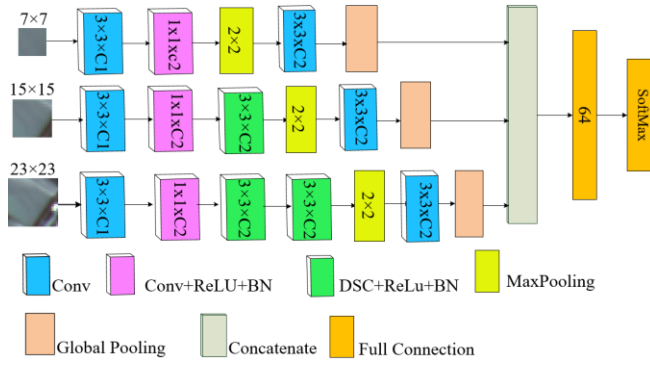


Figure 1. the designed lightweight CNN model. The “DSC” means depthwise separable convolution and “Conv” means standard convolution, BN means batch normalization, C1 =32 and C2=64 represent the number of the kernel.

In **HRS** images, ground objects usually appear at different scales. Therefore, it is necessary to describe the multi-scale variability of spatial features in **HRS** images. To capture multi-scale features, we designed a multi-scale CNN with three parallel branches. The sizes of the input image patches of the three branches are 7×7, 15×15 and 23×23 respectively as shown in Fig.1. **The reason why these three specific scales of the patches are chosen as the inputs of the model is that a good compromise between accuracy and computation time can be achieved after many trials and errors.** The outputs of the three parallel branches is concatenated to combine the multiscale features and then fed into a full-connection layer with 64 neurons. Note that the ReLU is used as the activation function for each convolution layer and deep separable convolution layer, then performs normalization, except for the first and the last convolution layer of each branch. The last one is the classification layer with softmax as the nonlinear activation function.

Generally, the multi-branch network contains more convolutional layers than the single-branch network, which results in more parameters. To reduce the number of parameters and the consumption of computational resources, we replace part of standard convolution with deep separable convolution. **The model is built from scratch with fewer parameters (1.1×10^5), lighter model (1.2 MB) and higher efficiency (2.8×10^4 patches/s).**

2.1. Datasets

Two sub-regions of Beijing and Qingdao, China, were chosen as the study areas. These two regions are strongly heterogeneous with a mixture of urban surface (e.g., asphalt, concrete) and semi-natural environment (e.g., vegetation, bare soil), therefore representing a good test for classification algorithms.

The Two scenes of GaoFen2 imagery of Beijing and Qingdao were captured on 5 September 2016 and 28 November 2017, with 4m spatial resolution and four multispectral bands (Red, Green, Blue, and Near Infrared).

The two sub-regions of Beijing (4214×4268 pixels) and Qingdao (3800×3782 pixels) were selected to investigate the effectiveness of the proposed CNN model. There were five land cover categories in the two study areas, including impervious surface (IS), bare soil, vegetation, water, and shadow. Sample points were collected using a stratified random scheme with the assistance of Google Earth images and GaoFen2 imagery. Then all samples for each class were split into 70% for training and 30% for testing (Table 1).

Table 1 Number of samples for training and testing

Study area	Class	Train	Test
Beijing	IS	447	192
	soil	385	145
	vegetation	405	173
	water	421	185
Qingdao	IS	260	112
	soil	243	104
	vegetation	320	127
	water	225	96

3. RESULTS AND DISCUSSIONS

3.1 Experiment setup

The performance of the proposed model was evaluated in two regions (Beijing and Qingdao) using two GaoFen2 multispectral images. **Our experiments are performed on a computer with Intel (R) i7-9700 K CPU, NVIDIA RTX1050Ti GPU, and 16G memory. We use CUDA 10.0 and cuDNN 7.6 to call GPU for training acceleration.** In the training process, the learning rate is set to 0.001, the batch size is set to 32, and train 20 epochs using the Adam optimizer. All experiments were conducted using the Keras API with TensorFlow as the backend.

Meanwhile, to verify the performance of the proposed model for **HRS** imagery classification, we compared it with traditional classification methods Random Forest [14]. For the Random Forest, the four bands of the GaoFen2 images and the eight texture features of mean, variance, homogeneity, contrast, dissimilarity, entropy, second moment and correlation were selected as feature variables. The addition of texture features obtained from the gray-level co-occurrence matrix [15], which enabled Random Forest to also utilize the spatial information of the image. Besides, in order to prove the effectiveness of the multi-branch structure, we also used the single-branch CNN with 15×15 image patches as inputs to compare with the multi-scale CNN. For a fair comparison, all methods were trained and tested on the same training and test sets and were tested using the overall accuracy (OA), kappa coefficient (Kappa), product accuracy (PA), and user accuracy (UA) of each class. The classification results

conducted by different methods were shown in Fig. 2 and 3 and in table 2 and 3. The details are given as below.

3.2 Analysis of Experimental Results

In this section, we evaluated the performance of the proposed method on two study areas. The classification results conducted by different methods were shown in Fig. 2 and 3 and in table 2 and 3. The details are given as below.

Table 2 Classification results of Beijing						
Class	Proposed		Single-branch CNN		Random Forest	
	PA%	UA%	PA%	UA%	PA%	UA%
IS	90.40	84.21	77.14	85.26	80.02	75.53
soil	87.58	84.28	96.12	62.26	65.02	76.65
vegetation	87.70	98.21	87.78	94.61	79.52	61.34
water	93.57	91.43	91.17	88.57	65.40	83.97
shadow	91.92	93.33	82.51	94.35	46.96	74.09
OA%	90.36		84.56		70.38	
Kappa	0.84		0.81		0.65	

Table 3 Classification results of Qingdao						
Class	Proposed		Single-branch CNN		Random Forest	
	PA%	UA%	PA%	UA%	PA%	UA%
IS	80.87	82.30	72.11	66.10	68.12	71.53
soil	85.39	85.41	79.59	82.67	86.02	76.35
vegetation	94.89	92.19	90.18	92.65	91.52	82.34
water	96.20	86.36	89.42	91.18	85.40	90.82
shadow	89.91	96.37	89.26	89.09	52.36	84.09
OA%	89.64		80.13		80.72	
Kappa	0.86		0.71		0.72	

From tables 2 and 3, the proposed model achieved the highest accuracy in most classes, and OA and Kappa were all the best values. From table 2 we can see that compared with single-branch CNN, the proposed model increases by 3.51% and 0.4% in OA and Kappa index, respectively. It demonstrates that the proposed model can use the multi-branch structure to extract multiscale information resulting in significantly improving classification accuracy. Notably compared with Random Forest, the classification accuracy of the CNN-based method (whether single-branch or multi-branch) is significantly higher in the most classes. Similar phenomena also appear in table 3. These results show the

effectiveness and robustness of the proposed method in terms of classification accuracy.

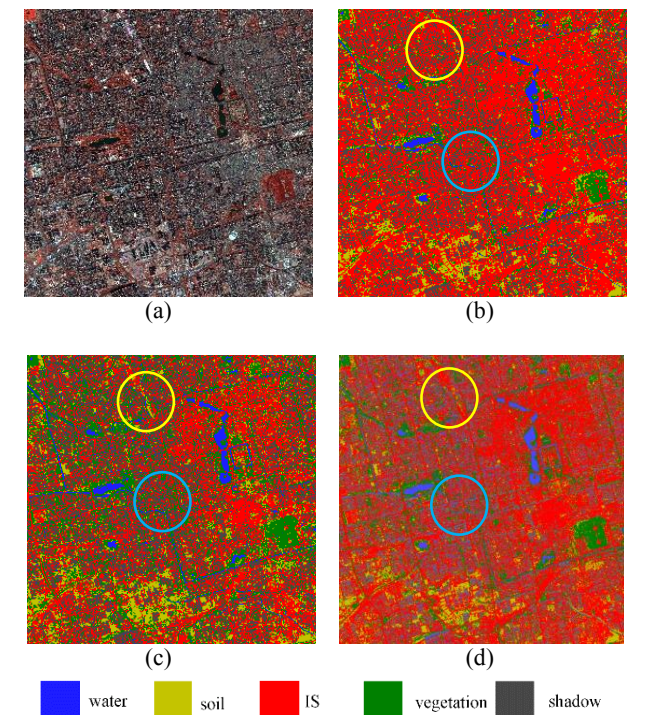


Fig. 2: The classification maps of the GaoFen2 multispectral image in Beijing: (a) False color image. (b) Proposed. (c) Single-branch CNN. (d) Random Forest.

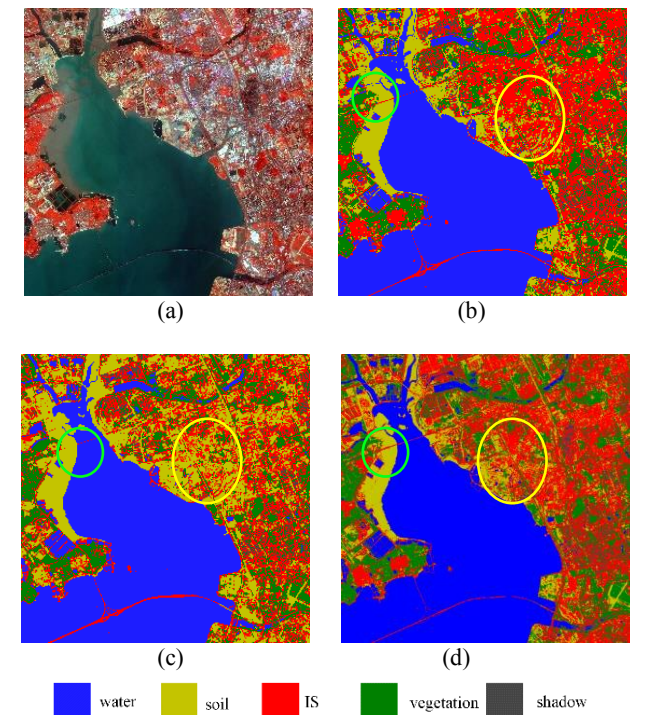


Fig. 3: The classification maps of the GaoFen2 multispectral image in Qingdao: (a) False color image. (b) Proposed. (c) Single-branch CNN. (d) Random Forest.

The classification maps are shown in Figs.2 and 3 also indicate the superiorities of the proposed model. From Fig. 2 (b) (c) and (d), it can be seen that compared with single-branch CNN and Random Forest, the proposed model can significantly reduce the misclassification phenomenon. Although the classification maps of single-branch CNN and multi-branch CNN are similar, fewer impervious surface pixels are wrongly classified as bare soil in the multi-branch CNN (marked in yellow ellipses). Compared with Random Forest, the proposed model significantly reduced the confusion between shadow and water (marked in blue ellipses). Similar phenomena also can be seen in Fig.3. Besides, as shown in Fig.3 single-branch CNN miss more small objects (highlight in green ellipses), which further proves that multi-scale information can enhance the performance of **HRS** images classification.

Among these three methods, our proposed method obtained the best classification results. Overall, the proposed model can be employed as a promising method for **HRS** images classification.

4. CONCLUSIONS

In this article, we proposed a lightweight and multi-scale CNN model for accurate and timely **HRS** images classification. It consists of three parallel branches, each of which inputs image blocks of different scales. The multiscale CNN architecture is very useful to capture multi-scale information in **HRS** images to improve the classification accuracy. Besides, to reduce the additional parameter increase caused by the multi-branching structure, we replace the part of the standard convolution with the deep separable convolution. The number of parameters in the whole model is only 1.1×10^5 . The proposed model was evaluated in two regions using GaoFen2 images, and compared with other state-of-the-art methods. The experiment results proved that the proposed model can boost the performance of **HRS** images land-cover classification.

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